Numbers in Colonial Maya Texts

Alex LaPrevotte

Research Questions:

- How frequently were Spanish numbers, Maya numbers, and numerals used in colonial-era Maya-language texts, and were they used interchangably?
- Did numeral classifiers function the same way in earlier days of Maya-Spanish contact that they do now?

Introduction: Numeral Classifiers

Numeral classifiers are words or affixes appended to numerals, which express characteristics of that numeral.

- Sortal classifiers
- Mensural classifiers

Numeral classifiers in modern Maya are often described as “obligatory” (Butler, Macri), but analysis of usage indicates that this is only the case when the number is Maya in origin.

In modern Maya:

Maya Numbers:			Maya classifiers:			Classified Numbers:
			p'éel	**inanimate objects**		
jun	'one'		*túul*	people, animals, gods		*junp'éel*
ka'a	'two'		*kúul*	plants, hair		*ka'ap'éel*
óox	'three'		*ts'íit*	long, thin objects		*óoxp'éel*
kan/kwaatro	'four'		*téen*	times		*kwaatro u p'éelel*
jo'o/siinko	'five'	+ sortal	*wáal*	round, flat things	=	*siinko u p'éelel*
seeys	'six'		*xéet'*	slices		*seys u p'éelel*
syeete	'seven'		*jéek*	pieces, branches		*syeete u p'éelel*
oocho	'eight'		*kúuch*	loads		*oocho u p'éelel*
nweebe	'nine'	mensural	*múut'*	boxes		*nweebe u p'éelel*
dyees	'ten'		*wúuts'*	kilograms		*dyees u p'éelel*

Numeral classifiers are not obligatory with Spanish numbers.

Y yo dicho interprete gen[era]l lo hise a todo mi leal saber y en tender y lo firme oy en veinte de nobiembre de mil seteçientos y onse años = Nicolas cardeña

Doc. 63, p. 77; Roys. 207, p. 308[179]

01

…

...

...

...

...

02

… 88 lay kin …

... 88 lay k'iin ...

... 88 lay k'iin ...

... 88 este día ...

... 1788 este día ...

03

… u Justisia y Regi[dores] ¥ escr[ibano] …

... ujustisya yéetel regidores yéetel eskribano

... u- justisyas yéetel rejidores yéetel eskribano

... 3A- justicias con regidor con escribano

... los justicias y los regidores y el escribano

04

… [Eb]tun ¥ tu uichil u cahalob u mult[unil]…

... Ebtun yéetel tuichil u kajalo'ob umultunil ...

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[179] Hay una copia de este documento en el doc. 73, p. 84.

353

Let's talk about the data

- In 2025, Dr. Julien Machault published his dissertation, which included the digitization, analysis, and translation of many Maya and Spanish documents written between 1561-1869.
- Over 1500 pages of his dissertation are data. I worked with a sample I extracted of 200 pages.
- This data includes Maya text, broken into a five-line format for analysis:
  - a transcription of the original colonial Maya text (in original orthography)
  - a transcription of the text with modern orthography
  - a morphological breakdown
  - a morphological gloss
  - a Spanish translation
- The data also include pieces of all-Spanish text, numbers, document names, and footnotes.

So, my challenge was to read this data in, get it into a workable format, find the numbers, and then look at how they are used and where they are paired with classifiers.

Y yo dicho interprete gen[era]l lo hise a todo mi leal saber y en
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[179] Hay una copia de este documento en el doc. 73, p. 84.

353

Data read-in and cleaning process

- Used pdftools to read the data and created a vector where each line of the PDF was a line of data
- Combined morphological breakdows into single tokens

la'	-e'	ti'	u-	k'ul	nah	aj-	tepal	yéetel
PROX	-TOP	PREP	3A-	sagrado	casa	AG-	gobierno	con

la'-e'	ti'	u-k'ul	nah	aj-tepal	yéetel
PROX-TOP	PREP	3A-sagrado	casa	AG-gobierno	con

Y yo dicho interprete gen[era]l lo hise a todo mi leal saber y en tender y lo firme oy en veinte de nobiembre de mil seteçientos y onse años = Nicolas cardeña

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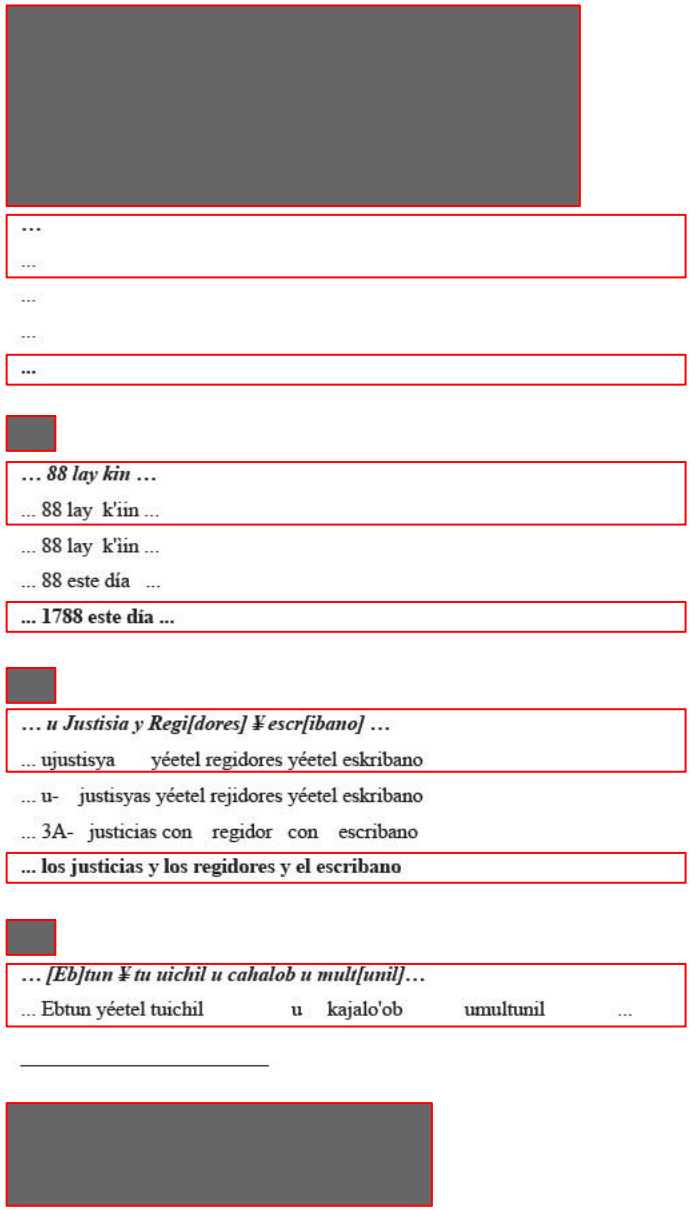
[179] Hay una copia de este documento en el doc. 73, p. 84.

353

Data read-in and cleaning process

- Converted everything to lower case
- Manually eliminated
  - incomplete chunks
  - parts entirely in Spanish
  - footnotes
  - chunks where one or more lines goes onto a second row, disrupting the 5-line chunk format
- Eliminated rows that only include numbers, and spaces as well as rows that start with "doc.", "verso", or "sigue en"

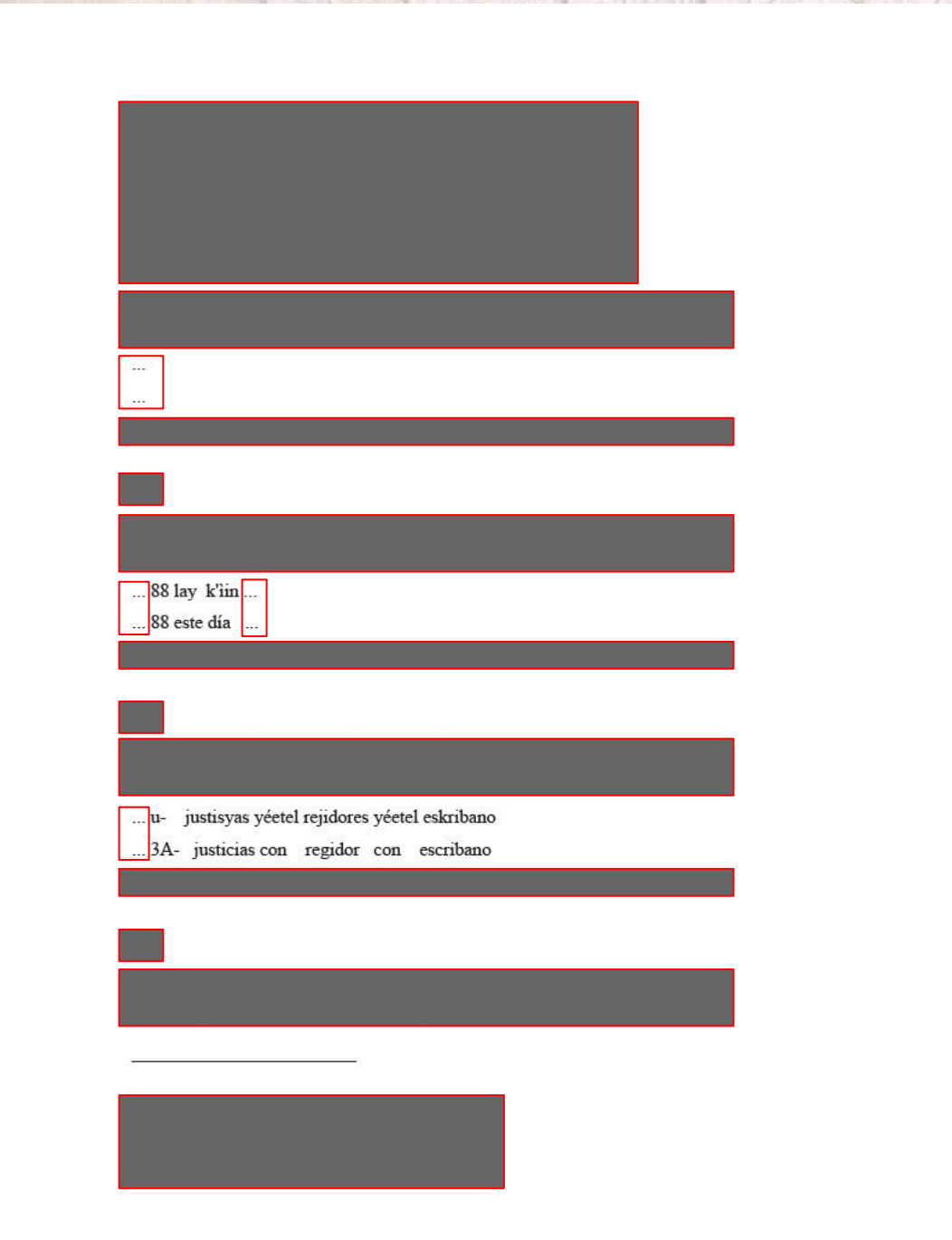


Data read-in and cleaning process

- Broke the data into 5-line “chunks”
- Eliminated rows 1, 2, and 5
- Broke each chunk down into words
- Eliminated each chunk with NAs

junmol	-o'on	ichil	audyensya
juntar	-1pl	adentroaudiencia	

- Concatenated all rows by chunk, giving me two long rows, and then turned those rows into columns:
  - morphological breakdown
  - morphological gloss

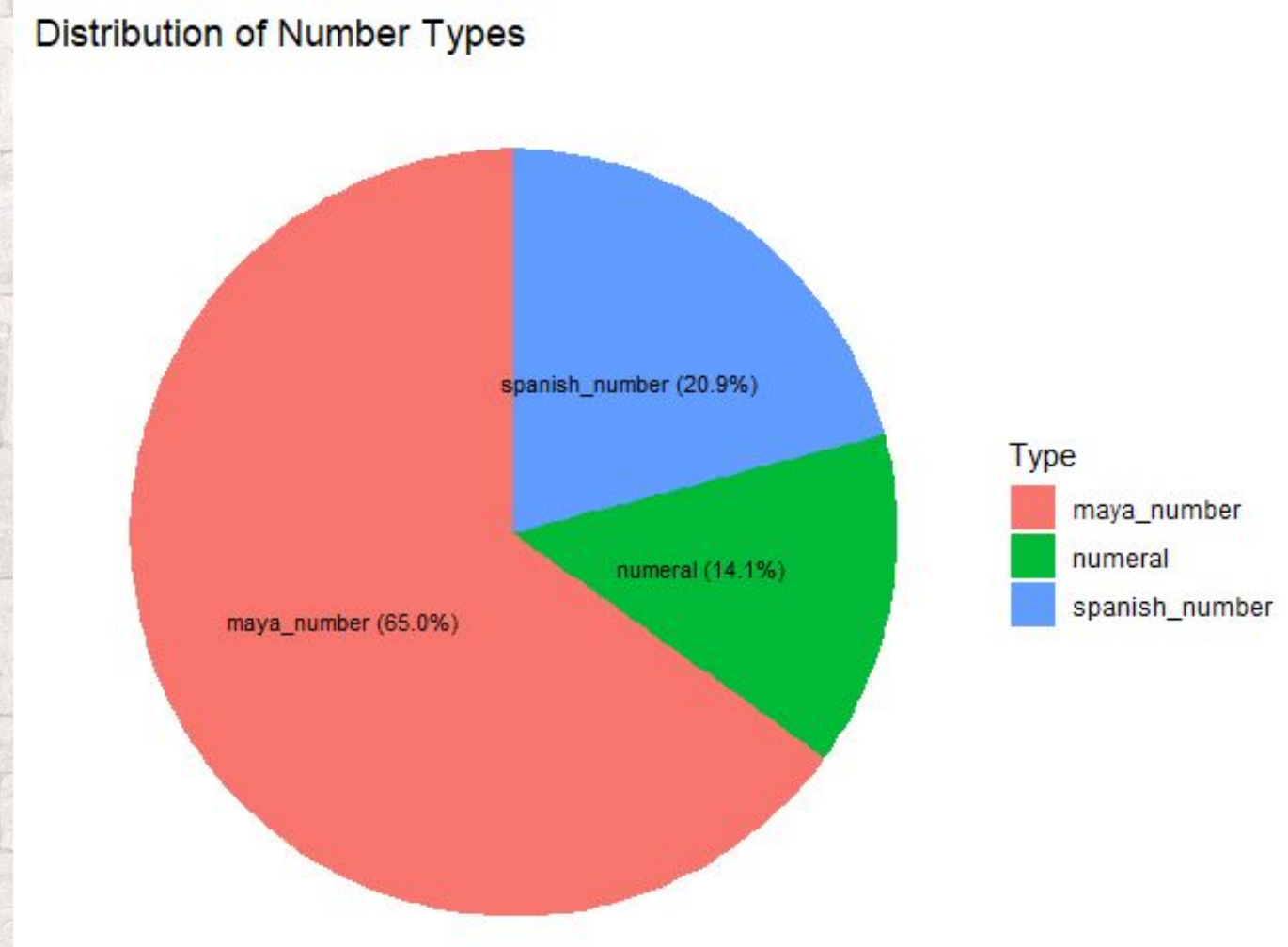


Final clean-up

- Trimmed everything
- Removed cells that were just ellipses
- Named my columns

This left me with 2 columns of 3989 rows. This was the data with which I moved forward.

type <chr>	count <int>
numeral	23
maya_number	106
spanish_number	34

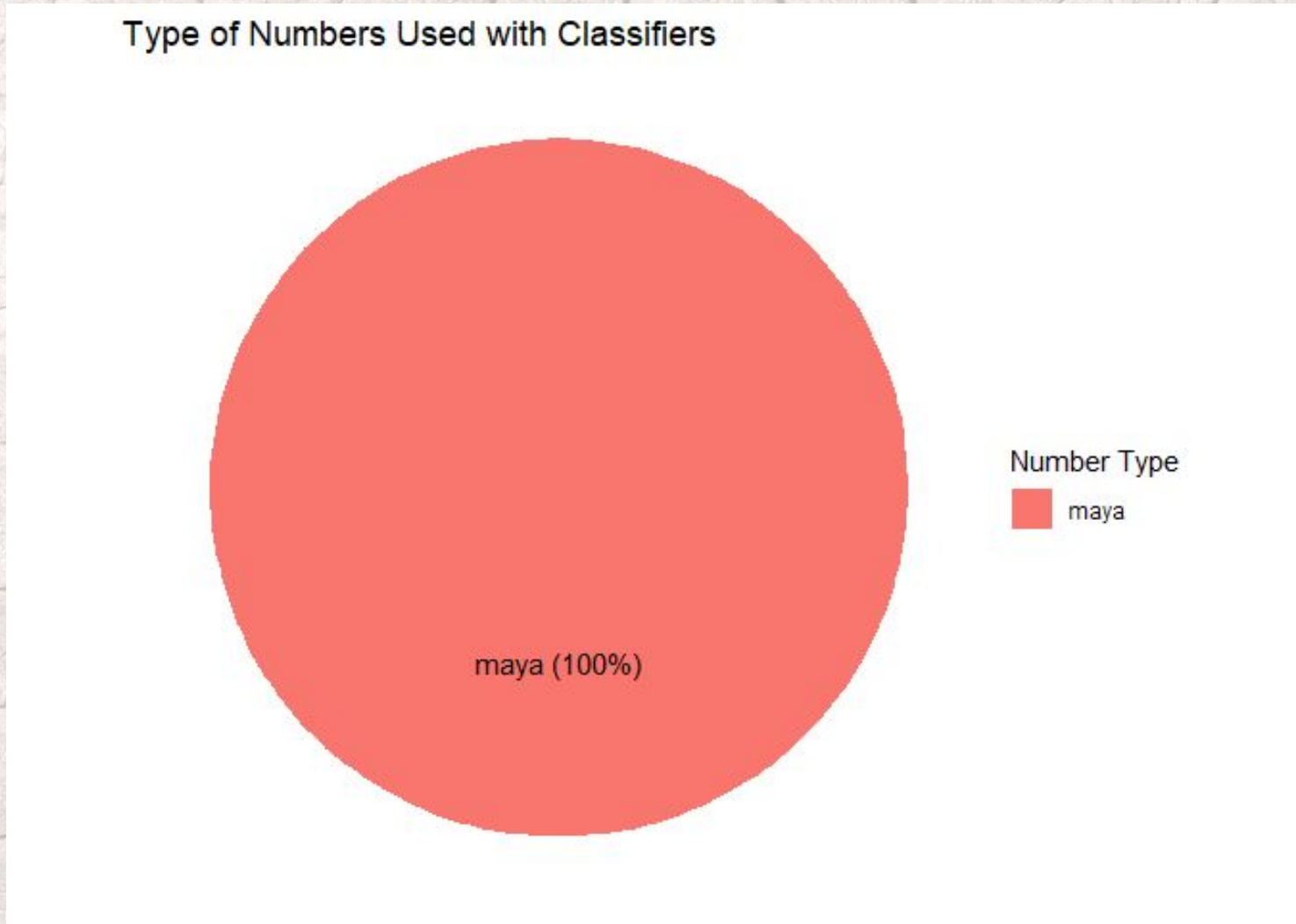


Analysis

- I wanted to look at numbers, so first I had to find the numbers. I read in lists I created of Spanish and Maya numbers and also searched for numerals.
  - I normalized everything for accents.
- To qualify as a number for my analysis, the number had to be present in both the breakdown and gloss columns.

Analysis

- In modern Maya, numeral classifiers are usually either affixed to numbers or come two words later (*e.g.* seeys u p'éelel), so, to be safe, I created 4-grams where the first token was each number identified in the analysis, and then I searched through those tokens and the three that followed, looking for classifiers.
- I based my search on a list of classifiers that are common in modern Maya.
- I also searched for morphemes glossed as "CN," which was how many classifiers in the data were glossed. This helped me identify three additional classifiers that are no longer in common use, which I added to my search terms.



Surprise! 100% of the classifiers I was able to identify in the text were paired with Maya-language numbers.

Analysis

Why might that be? Let's look at the numeral and Spanish 4-grams:

index	type	b1	b2	b3	b4
153	numeral	1600	'àanyo-s	tak-ts'a'-aj-	u-jaj-il
860	numeral	88	lay	k'ìin	u-justisyas
972	numeral	8	de	mayo	de
976	numeral	1791	'àanyo-s	ts'a'-ik-	u-jaj-il
1101	numeral	1711	don	francisco	xul
1147	numeral	30	de	enero	de
1151	numeral	1713	'àanyos	don	pedro
1334	numeral	1711	teen	don	diego
1502	numeral	8	de	nobyembre	de
1756	numeral	23	de	julyo	de
2206	numeral	30	de	enero	de
2210	numeral	1713	'àanyos	bin-el	kaj
2293	numeral	30	de	enero	de
2297	numeral	1713	'àanyo-s	way	ti'
2365	numeral	20	josef	camal	escribano
2371	numeral	30	de	abril	de
2375	numeral	1713	'àanyo-s	bey	xan
2427	numeral	40	yéetel	ka'a-k'an	yéetel
3024	numeral	19	de	abril	de
3028	numeral	1714	'àanyos	lae'	don
3396	numeral	31	de	mayo	mil
3669	numeral	31	de	mayo	de
3817	numeral	1758	to'on	k--o'on	batab

index	type	b1	b2	b3	b4
434	spanish	mil	sèeys	syentos	juan
435	spanish	sèeys	syentos	juan	dzul
436	spanish	syentos	juan	dzul	tekom
712	spanish	syentos	'àanyos	don	fabian
739	spanish	mil	setesyentos	ti'	junmol-o'on
740	spanish	setesyentos	ti'	junmol-o'on	way
1018	spanish	onse	andres	maste	k-in-ts'a'-ik-
1376	spanish	oocho	de	nobyembre	de
1380	spanish	mil	setesyentos	y	onse
1381	spanish	setesyentos	y	onse	teen
1383	spanish	onse	teen	andres	maste
1506	spanish	mil	setesyentos	y	onse
1507	spanish	setesyentos	y	onse	don
1509	spanish	onse	don	francisco	xul
1760	spanish	mil	setesyentos	sesenta	y
1761	spanish	setesyentos	sesenta	y	uno
1762	spanish	sesenta	y	uno	capitán
1764	spanish	uno	capitán	don	salvador
2363	spanish	dosyentos	y	20	josef
3256	spanish	mil	setesyentos	y	dyèes
3257	spanish	setesyentos	y	dyèes	y
3259	spanish	dyèes	y	syèete	'àanyo-s
3261	spanish	syèete	'àanyo-s	don	francisco
3399	spanish	mil	setesyentos	y	kwarenta
3400	spanish	setesyentos	y	kwarenta	y
3402	spanish	kwarenta	y	trèes	'àanyos
3404	spanish	trèes	'àanyos	capitán	don
3673	spanish	mil	setesyentos	y	kwarenta
3674	spanish	setesyentos	y	kwarenta	y
3676	spanish	kwarenta	y	capitán	don
3885	spanish	mil	setesyentos	setenta	y
3886	spanish	setesyentos	setenta	y	ocho
3887	spanish	setenta	y	ocho	ka'
3889	spanish	ocho	ka'	t-u-kon-aj-	in-yuum

Analysis

Ah, here is the same list with all of the dates removed

860	numeral	88	lay	k'ìin	u-justisyas
2365	numeral	20	josef	camal	escribano
2427	numeral	40	yéetel	ka'a-k'an	yéetel

1018	spanish	onse	andres	maste	k-in-ts'a'-ik-
2363	spanish	dosyentos	y	20	josef

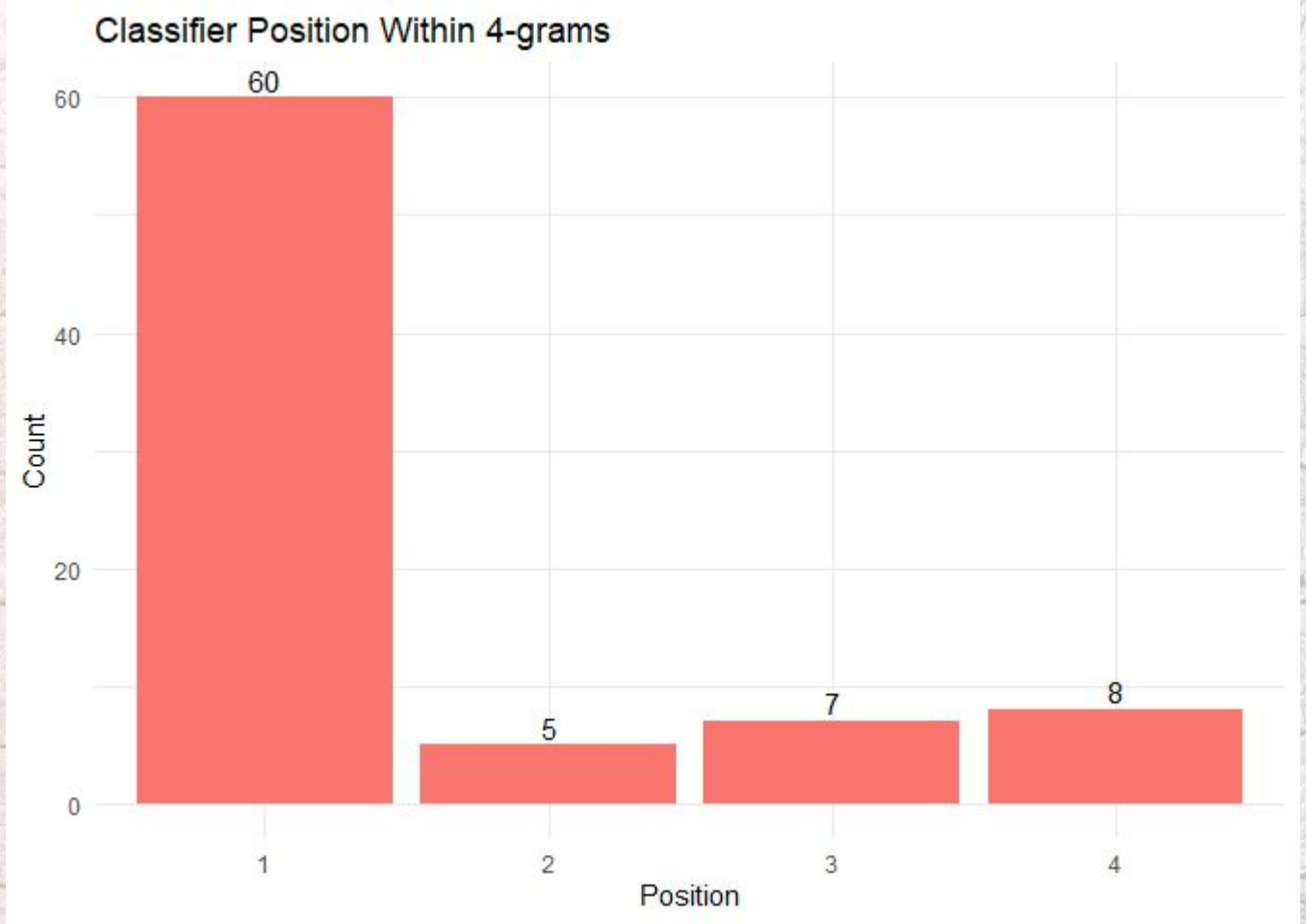
Analysis

So, it seems like Spanish numbers and numerals were primarily used for dates, at the time these documents were written. There were a few instances in the Maya data where numeral classifiers appeared to be paired with dates (more often with numbers of days and years), but it's infrequent and it makes sense that they wouldn't be used for that purpose, since the Maya used a totally different and incompatible timekeeping system.

It is also noteworthy, regarding the infrequency of classifiers co-occurring with numerals, that in classic Maya glyphic writing, numbers were always written numerically, not spelled phonetically, and there is no concrete evidence of numeral classifiers used in Maya epigraphs, so they may have just seemed unnecessary in that form to the Maya.

Analysis

Looking at just the Maya data, now, where do the classifiers occur?



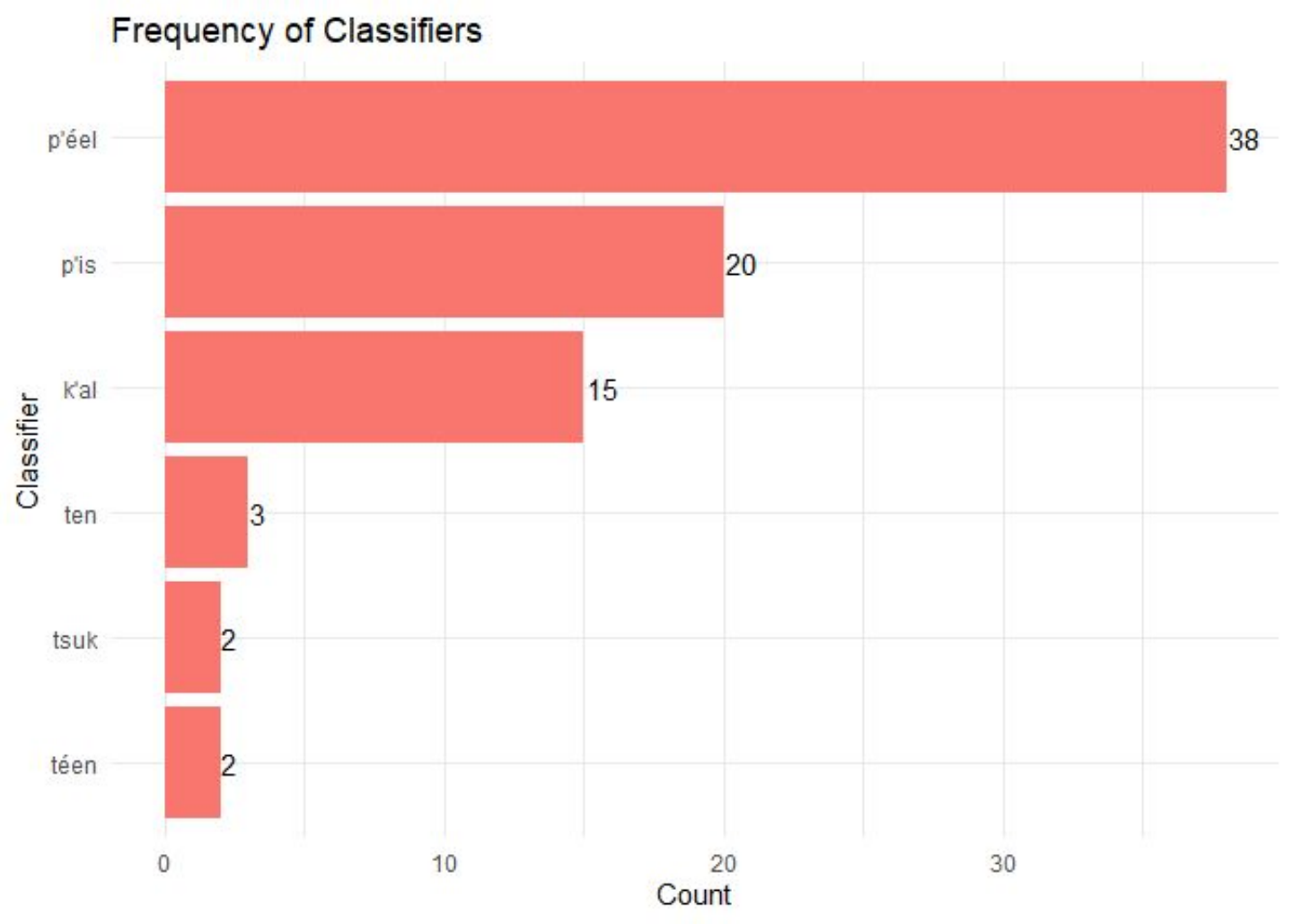
- Overwhelmingly in the N1 position, affixed to the number
- Some of the later appearances are also appended to the number, and the number is simply multiple words, therefore showing up multiple 4-grams in a row

óox-	k'al	peso	ka'tak	jo'lajun-p'is

NUMBER CLASSIFIER

Analysis

I also did a little investigation of how often each classifier was used.



- p’éel - inanimate objects
- p’is - mensural, mostly numbers of days or years
- k’al - bundles or batches
- ten/téen - times
- tsuk - places

Research Questions:

How frequently were Spanish numbers, Maya numbers, and numerals used in colonial-era texts, and were the used interchangably?

- Maya numbers were used the most by far. It also appears Spanish Numbers were used slightly more frequently than numerals, but Maya and Spanish numbers are both over-represented, so those could be closer than the data indicates.
- They were not interchangeable, numerals and Spanish numbers were primarily used for dates, while Maya numbers were used in varied applications.

Research Questions:

Did numeral classifiers function the same way in earlier days of Maya-Spanish contact that they do now?

- They were overwhelmingly used appended to Maya numbers, which is still true, but it appears they were not used with Spanish numbers or numerals at all in this data, and there were also a few organic instances of numeral classifiers as separate words following Maya numbers, which is uncommon now.

¡Problems

- Having to clean things like footnotes and Spanish-language portions out of the data manually is labor-intensive and makes for brittle code.
- Several Maya and Spanish numbers are multiple words long or occur near other numbers, and are therefore over-represented in the data.

Future Research!

- I would love to try to apply this analysis to the whole data set.
- This pipeline, with minor tweaking, could be applied to other similarly-formatted data.

Yáan waaj k’áatchi.

Do you have any questions?

References:

Butler, L. K. (2011). The Morphosyntax and Processing of Number Marking in Yucatec Maya (dissertation).

Julien Machault. *Los títulos de ebtún : transcripción, traducción y análisis histórico*. Ph.D. Dissertation, Universidad Nacional Autónoma de México (UNAM), 2025.

Macri, M. (2000). Numeral Classifiers and Counted Nouns in the Classic Maya Inscriptions. Written Language & Literacy. 3. 13-36.